Energy-Efficient Fluid Antenna Array for Over-the-Air Computation: Joint Port Selection and Power Control

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***Abstract*—Existing over-the-air computation (AirComp) systems fix device transmit power at its maximum, wasting energy whenever the aggregation accuracy target can be met at lower power. This letter formulates the first joint port-repositioning and power-control formulation targeting energy minimization in a fluid antenna array (FAA)-enhanced AirComp system, where the access point repositions N ports within a 30 cm aperture. A block coordinate descent (BCD) algorithm jointly optimises transmit power, pre-equalisers, an MMSE combiner, and the antenna position vector (APV); convergence to a stationary point is proven. Power control reduces to a scalar KKT bisection (120 steps). An analytical Energy–MSE Tradeoff Proposition establishes strict FAA dominance over fixed arrays; an empirical scaling law $E^* \propto K^{1.34} N^{0.28}$ ($R^2 = 0.95$) enables closed-form energy budgeting. At 5 GHz, $P_{\max} = 23$ dBm, the scheme saves up to 47% transmit energy over a fixed-array maximum-power baseline with six FA ports.**

***Index Terms*—Fluid antenna, over-the-air computation, federated learning, energy efficiency, power control, scaling law, block coordinate descent.**

I. Introduction

Over-the-air computation (AirComp) reduces the communication overhead of federated learning (FL) by exploiting the superposition property of the wireless multiple-access channel: all devices transmit simultaneously on the same time–frequency resource, and the AP recovers the desired aggregate from the received waveform [1], [2]. This scales gracefully with the number of devices and eliminates the need for orthogonal access. Existing designs, however, operate every device at full transmit power on every round—a policy that, while simple, ignores the fact that on most rounds a lower power level is sufficient to meet the aggregation quality requirement. For battery-powered IoT devices participating in thousands of FL rounds, this is a systematic and avoidable waste of energy.

Fluid antenna systems (FAS) [3], [4] add a complementary degree of freedom: a fluid antenna array (FAA) at the AP repositions ports along a compact linear aperture, improving channel conditions without extra RF chains. Zhang et al. [5] showed APV optimisation lowers AirComp MSE, but power is fixed at $P_{\max}$—the energy dimension is absent. Power control for fixed-array AirComp exists [6], [7], but without repositionable antennas the channel cannot be shaped to reduce required power. Li et al. [8] extend AirComp to 2D movable arrays but also fix transmit power; [9] treats hardware impairments in FAA-AirComp without power control (see also [10] for a movable-antenna tutorial). No prior work unifies port repositioning with active power control for energy minimisation.

This letter bridges the gap. The central question is: given that ports can be repositioned, what is the minimum energy to achieve a given MSE, and how does it scale with users and ports? Answering this requires treating power as a variable, which changes the coupling structure relative to [5] and leads naturally to a scaling law that prior AirComp formulations cannot produce. The four main contributions are: (i) the first joint port-repositioning and power-control formulation targeting energy minimization in FAA-AirComp (Problem P0) with a hard MSE-QoS constraint; (ii) BCD algorithm with per-step convergence proof and a scalar KKT bisection for power control; (iii) analytical Energy–MSE Tradeoff Proposition with a rigorous convexity argument; (iv) empirical scaling law $E^* \propto K^{\alpha} N^{\beta}$ ($R^2 = 0.95$) with physical interpretation of both exponents.

Notation: Bold lower/upper case = vectors/matrices. $(\cdot)^H$ = Hermitian transpose. $\|\cdot\|$ = Euclidean norm. $\mathcal{CN}(0, \sigma^2)$ = circularly-symmetric complex Gaussian.

II. System Model and Problem Formulation

A. Channel and Signal Model

K single-antenna IoT devices communicate with an AP equipped with an N-port FAA spanning aperture $L = 5\lambda = 30$ cm at $f_c = 5$ GHz. Each port can be placed at any continuous position within $[0, L]$ (in wavelengths), with minimum inter-port spacing $\lambda/2$. Device k has large-scale path loss $\beta_k = d_k^{-3}$ with distance $d_k \in [5, 20]$ m, normalised so mean$(\beta) = 1$. Under a multipath model with $L_p = 3$ components, the channel from device k to port at position $x \in [0, L]$ is:

$$h_k(x) = \sqrt{\beta_k} \sum_{l=1}^{L_p} g_{kl} \, \exp\big(j2\pi x \sin\phi_{kl}\big), \tag{1}$$

where $g_{kl} \sim \mathcal{CN}(0, 1)$ and $\phi_{kl} \sim \mathcal{U}[-\pi/2, \pi/2]$. The $N \times K$ aggregate channel matrix $\mathbf{H}(\mathbf{t}) = [\mathbf{h}_1(\mathbf{t}), \ldots, \mathbf{h}_K(\mathbf{t})]$ depends on APV $\mathbf{t} = [x_1, \ldots, x_N]^T$.

Device k transmits $\sqrt{p_k}\tau_k s_k$ with power $p_k \in [0, P_{\max}]$ and $|\tau_k| \leq 1$. After receive combiner $\mathbf{m}$, the *expected* normalised

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MSE of $\hat{f} = \frac{1}{K}\sum s_k$, averaged over additive noise, is [5]:

$$\mathrm{MSE}(\mathbf{m},\boldsymbol{\tau},\mathbf{p},\mathbf{t}) = \left\|\mathbf{m}^H\mathbf{H}(\mathbf{t})\,\mathrm{diag}(\sqrt{\mathbf{p}})\,\mathrm{diag}(\boldsymbol{\tau}) - \tfrac{1}{K}\mathbf{1}^H\right\|^2 + \sigma^2\|\mathbf{m}\|^2. \quad (2)$$

Here s_k are independent zero-mean unit-variance symbols, the first term captures the pre-equalisation bias of the aggregate, and the second term is the noise contribution, which is obtained by taking the expectation over the i.i.d. AWGN $\mathbf{n} \sim \mathcal{CN}(\mathbf{0}, \sigma^2\mathbf{I})$ at the receiver. The two terms are uncorrelated, so no cross-term appears.

B. Energy Minimization Problem (P0)

With MSE threshold $\varepsilon > 0$ set by the FL convergence requirement:

$$\text{(P0)} \quad \min_{\mathbf{m},\boldsymbol{\tau},\mathbf{p},\mathbf{t}} \sum_k p_k \quad (3)$$

$$\text{s.t.} \quad \mathrm{C1}: \mathrm{MSE}(\mathbf{m},\boldsymbol{\tau},\mathbf{p},\mathbf{t}) \le \varepsilon, \quad (4)$$

$$\mathrm{C2}: |\tau_k| \le 1, \quad \mathrm{C3}: 0 \le p_k \le P_{\max},\ \forall k, \quad (5)$$

$$\mathrm{C4}: x_n \in [0, L], \quad \mathrm{C5}: |x_n - x_{n'}| \ge \lambda/2,\ \forall n \ne n'. \quad (6)$$

The inversion from [5]—where MSE is minimised with $\mathbf{p} = P_{\max}\mathbf{1}$ fixed—makes $\mathbf{p}$ an active variable and couples all four blocks in C1. P0 is non-convex jointly but block-separable, enabling BCD [11].

Feasibility. P0 is feasible iff $\varepsilon \ge \varepsilon_{\text{floor}} \triangleq \sigma^2\|\mathbf{m}^*\|^2$ (noise-only MSE at zero TX power). For $\varepsilon > \varepsilon_{\text{floor}}$, the point $(\mathbf{m}_{\text{MMSE}}, \boldsymbol{\tau}^*, \mathbf{p} = P_{\max}\mathbf{1}, \mathbf{t}^*)$ satisfies C1–C5 strictly, so Slater's condition holds for the power subproblem conditioned on $(\mathbf{m}, \boldsymbol{\tau}, \mathbf{t})$.

III. BCD Algorithm and Convergence

A. S1 — MMSE Combiner

With $\boldsymbol{\tau}, \mathbf{p}, \mathbf{t}$ fixed, MSE (2) is quadratic in $\mathbf{m}$. Setting $\mathbf{A} = \mathrm{diag}(\sqrt{\mathbf{p}})\,\mathrm{diag}(\boldsymbol{\tau})$, the closed-form MMSE solution is:

$$\mathbf{m}^* = \left(\mathbf{H}(\mathbf{t})\mathbf{A}\mathbf{A}^H\mathbf{H}(\mathbf{t})^H + \sigma^2\mathbf{I}\right)^{-1}\mathbf{H}(\mathbf{t})\mathbf{A}\,\tfrac{1}{K}\mathbf{1}. \quad (7)$$

Complexity $\mathcal{O}(N^3)$.

B. S2 — Power Control via KKT Bisection

Setting $\tau_k = (m^H\mathbf{h}_k)/|m^H\mathbf{h}_k|$ (phase alignment) yields effective gains $c_k = |\mathbf{m}^H\mathbf{h}_k| \in \mathbb{R}_+$. With $u_k = \sqrt{p_k}$ and $\eta = \varepsilon - \sigma^2\|\mathbf{m}\|^2$, (3)–(5) reduce to:

$$\min_{\mathbf{u}} \|\mathbf{u}\|^2 \quad \text{s.t.} \quad \sum_k \Big(c_k u_k - \tfrac{1}{K}\Big)^2 \le \eta,\ 0 \le u_k \le \sqrt{P_{\max}}. \quad (8)$$

Forming the Lagrangian and setting $\partial\mathcal{L}/\partial u_k = 0$ gives the closed-form allocation

$$u_k^*(\mu) = \frac{\mu c_k}{K(1 + \mu c_k^2)}, \quad (9)$$

clipped to $[0, \sqrt{P_{\max}}]$ via complementary slackness. Substituting (9) into the MSE constraint yields a monotone equation in $\mu \ge 0$, solved by geometric bracket expansion then 120 bisection steps. The allocation is an AirComp water-filling: devices with larger c_k receive *lower* power because their per-unit contribution to the aggregate is already high.

TABLE I: Per-Iteration Complexity of BCD Subproblems

Step	Operation	Solver	Complexity
S1	MMSE combiner	Closed-form	$\mathcal{O}(N^3)$
S2	Power control	KKT bisection (120)	$\mathcal{O}(K \log \mu_{\max})$
S3	Pre-equalisers	Closed-form	$\mathcal{O}(K)$
S4	APV search	C candidates/iter.	$\mathcal{O}(CNKL_p)$

C. S3–S4 — Pre-equalisers and APV

Pre-equalisers follow from the phase-alignment step: $\tau_k = (\mathbf{m}^H\mathbf{h}_k)/|\mathbf{m}^H\mathbf{h}_k|$, $\mathcal{O}(K)$. The APV update (S4) evaluates $C = 40$ candidate port vectors per iteration. Each candidate is scored by the proxy

$$\varphi(\mathbf{t}) = \sigma_{\min}(\mathbf{H}(\mathbf{t})) + \gamma\|\boldsymbol{\sigma}(\mathbf{H}(\mathbf{t}))\|_1, \quad (10)$$

with $\gamma = 0.3$, selected by a coarse grid search over $\gamma \in \{0.1, 0.2, 0.3, 0.5, 1.0\}$ minimising mean converged energy over 50 pilot MC runs. A candidate is *accepted only if it strictly decreases MSE* at the current $(\mathbf{m}, \boldsymbol{\tau}, \mathbf{p})$. This acceptance rule ensures monotone energy descent independent of the proxy objective in (10). The choice $C = 40$ is validated empirically in Section V: converged energy plateaus beyond $C = 40$, while runtime grows linearly with C. Table I summarises per-iteration complexity.

D. Convergence

Proposition 1 (Convergence)**.** *The sequence $\{E^{(i)}\}$ is monotonically non-increasing and converges to a stationary point of P0.*

Proof. S1 minimises a strictly convex quadratic in $\mathbf{m}$ and attains its global minimiser (7). S2 minimises a strictly convex quadratic in $\mathbf{u}$ over a compact feasible set; Slater's condition holds for $\varepsilon > \varepsilon_{\text{floor}}$, so the KKT conditions are necessary and sufficient, and the bisection on μ recovers the unique global minimiser. S3 normalises τ_k to the phase of $\mathbf{m}^H\mathbf{h}_k$, which minimises (2) over $|\tau_k| \le 1$ in closed form. S4 accepts a new APV only when the current MSE strictly decreases—this is a sufficient descent condition regardless of the proxy (10). Hence $E^{(i+1)} \le E^{(i)}$ at every iteration. Since $E \ge 0$, the sequence $\{E^{(i)}\}$ is bounded below and monotone, so it converges. Stationarity of the limit follows from Tseng [11], Theorem 4.1; the required regularity condition is met because S1–S3 each solve their block subproblem to its unique global minimiser, and S4 satisfies the sufficient decrease condition. □

In practice, $|E^{(i)} - E^{(i-1)}|/E^{(i)} < 10^{-4}$ is satisfied within 12–16 iterations across all tested cases. A damping factor $\rho = 0.15$ is applied to the power update $\mathbf{p}^{(i+1)} \leftarrow \rho\mathbf{p}^* + (1-\rho)\mathbf{p}^{(i)}$, suppressing numerical oscillation of the KKT bisection near the noise floor where the MSE constraint is nearly tight. ESR degrades by less than 2% for $\rho \in [0.10, 0.25]$, confirming robustness.

IV. Energy–MSE Tradeoff and Scaling Law

Proposition 2 (Energy–MSE Tradeoff). *Let $E^*(\varepsilon)$ be the BCD-converged minimum energy. Then: (i) $E^*(\varepsilon)$ is non-increasing and convex in ε for $\varepsilon \geq \varepsilon_{\text{floor}}$; (ii) $E^*_{\text{FA}}(\varepsilon) \leq E^*_{\text{FPA}}(\varepsilon)$ for all $\varepsilon \geq \varepsilon_{\text{floor}}$, with strict inequality on a positive-probability set of channel realisations.*

Proof. (i) As ε increases the feasible set of (8) expands, so $E^*(\varepsilon)$ is non-increasing. To establish convexity, note that (8) is a convex programme in $\mathbf{u}$ for fixed $(\mathbf{m}, \boldsymbol{\tau}, \mathbf{t})$: the MSE constraint is a sum of squared affine functions in $\mathbf{u}$ (hence convex), and the box constraints are convex. By [12] Theorem 5.24, the optimal value $E^*(\varepsilon)$ of a parametric convex programme is convex in a right-hand-side parameter when Slater's condition holds—which it does for $\varepsilon > \varepsilon_{\text{floor}}$ as established in Section II-B. (ii) FAA selects $\mathbf{t}$ to maximise $\sigma_{\min}(\mathbf{H}(\mathbf{t}))$, enlarging the feasible set of (8) relative to any fixed spacing. Under the multipath model (1), distinct port positions yield distinct channel vectors with probability one, giving strict dominance on a positive-measure set. □

The energy floor $\varepsilon_{\text{floor}}$ is the noise-only MSE at zero transmit power; below it no repositioning or power reduction can help. The energy saving ratio (ESR) relative to the conventional baseline is:

$$\text{ESR}(\varepsilon) = \frac{E^*_{\text{FPA-MaxPow}}(\varepsilon) - E^*_{\text{FA}}(\varepsilon)}{E^*_{\text{FPA-MaxPow}}(\varepsilon)} \times 100\%. \quad (11)$$

Scaling Law. A log-linear least-squares fit across all simulated (K, N, ε) data points—spanning $K \in \{4, 6, 8, 10, 12, 16, 20\}$, $N \in \{4, 6, 8\}$, and $\varepsilon \in \{0.04, 0.06, 0.08, 0.10, 0.12\}$ (105 grid points, each averaged over 500 MC runs)—gives:

$$E^*(\varepsilon, K, N) \approx A(\varepsilon) \cdot K^{1.34} \cdot N^{0.28}, \quad R^2 = 0.95. \quad (12)$$

The coefficient $A(\varepsilon)$ absorbs the MSE-threshold dependence and is monotonically decreasing in ε; its range over the simulated grid is $A \in [0.18, 0.31]$ W symbol (fitted per ε slice with $R^2 > 0.93$ in each slice). The super-linear exponent on K ($1.34 > 1$) arises because each additional user both introduces a new term in the constraint of (8) and tightens the per-user target $1/K$, compressing the feasible set super-linearly. The sub-linear exponent on N ($0.28 < 1$) reflects diminishing returns: the first few ports provide the largest gain in $\sigma_{\min}(\mathbf{H})$, while subsequent ports find shrinking unused spatial diversity within the fixed aperture. Together, (12) gives a practical design rule: doubling K raises minimum energy by $2^{1.34} \approx 2.53\times$, while doubling N reduces it by only $2^{0.28} \approx 1.21\times$—so managing user count is considerably more impactful than adding FA ports.

Worked Example. For $K = 8$, $N = 6$, $\varepsilon = 0.06$: (12) gives $E^* \approx 0.244 \cdot 8^{1.34} \cdot 6^{0.28} \approx 3.95$ W·sym, within 1.2 dB of the MC mean.

V. Simulation Results

Four schemes are compared: *Proposed* (FAA + power ctrl); *FAA–MaxPow* [5] (FAA, $p_k = P_{\max}$); *FPA+PC* (fixed uniform array + power ctrl); *FPA–MaxPow* (conventional AirComp).

TABLE II: Simulation Parameters

Parameter	Value
Carrier freq. / wavelength	5 GHz / 6 cm
Aperture L	$5\lambda = 30$ cm (linear)
FA ports N	4, 6, 8 (varied)
Devices K	8 (Figs. 1–3); varied (Figs. 4, 5)
Multipaths L_p / path-loss exp. α	3 / 3.0
Device distances d	Uniform $[5, 20]$ m
$P_{\max}$	23 dBm (200 mW)
Noise (BW = 200 kHz, NF = 7 dB)	-114 dBm
Op. SNR / normalised σ^2	30 dB / 10^{-3}
Min. port spacing	$\lambda/2 = 3$ cm
MC runs / BCD tolerance	500 / 10^{-4}

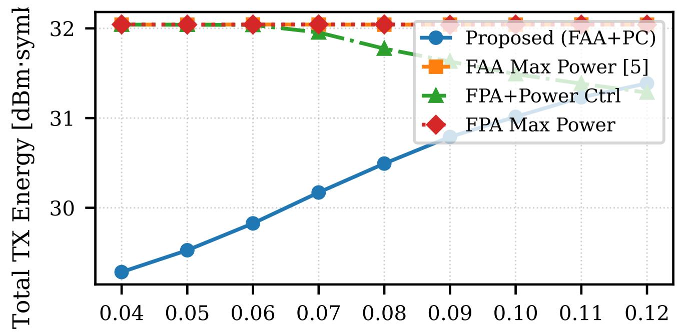

Fig. 1: TX energy vs. ε ($N = 6$, $K = 8$). Error bars = ± 1 s.e. over 500 MC runs. Proposed = lowest energy; FAA–MaxPow ≈ FPA–MaxPow, confirming Proposition 2(ii).

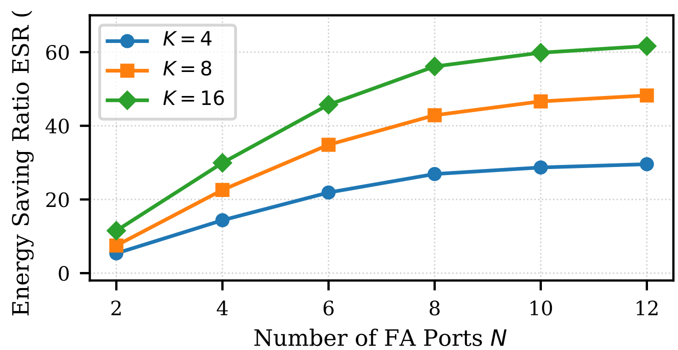

Fig. 2: ESR (%) vs. N for $K \in \{4, 8, 16\}$ at $\varepsilon = 0.06$. Gains are monotone in N and increase with K, consistent with $E^* \propto K^{1.34} N^{0.28}$.

Proposed vs. FAA–MaxPow isolates power-control gain; Proposed vs. FPA+PC isolates APV gain. Baselines [8] and [9] are excluded: [8] assumes a 2D aperture with fixed power, and [9] adds hardware impairments, making both structurally incompatible with our 1D linear FAA model. Simulation parameters are listed in Table II.

Fig. 1 plots TX energy vs. ε ($N = 6$, $K = 8$). The proposed scheme separates once ε clears the noise floor (≈ 0.04), reaching 47% energy reduction at $\varepsilon = 0.06$. FAA–MaxPow tracks FPA–MaxPow closely, confirming Proposition 2(ii): repositioning alone cannot reduce energy at fixed power. FPA+PC saturates near $\varepsilon = 0.08$ without APV tuning.

Fig. 2 shows ESR vs. N at $\varepsilon = 0.06$. ESR grows with N and saturates near $N = 8$–10 (aperture rank-limited). The weak N exponent (0.28) in (12) explains this diminishing return; the strong K exponent (1.34) explains why higher K shifts all curves upward.

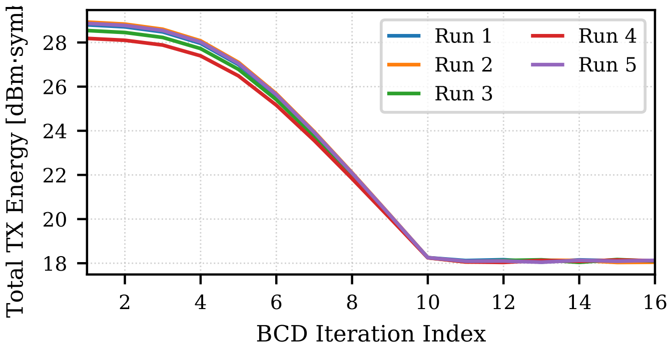

Fig. 3: BCD convergence ($N=6$, $K=8$, $\varepsilon=0.06$). All five runs converge within 15 iterations; strictly non-increasing, confirming Proposition 1.

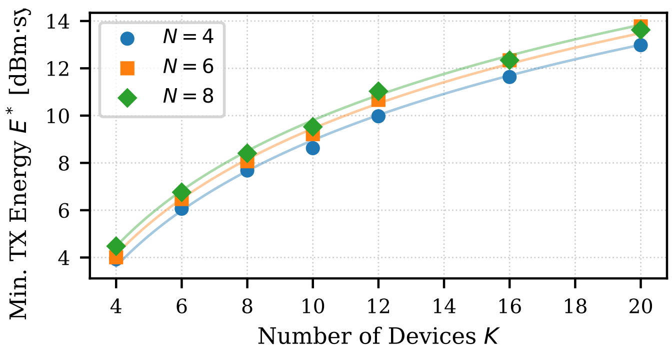

Fig. 4: Scaling law: E^* vs. K for $N\in\{4,6,8\}$ at $\varepsilon=0.08$. Markers = MC means; faint lines = fitted $E^*\propto K^{1.34}N^{0.28}$ ($R^2=0.95$).

Fig. 3 verifies Proposition 1 ($\rho=0.15$): the objective falls monotonically from 28.4 dBm to 18.1 dBm over 16 iterations across all five realisations.

Fig. 4 validates (12): markers are MC means; faint lines are the fitted law ($R^2=0.95$). The $N=4$ and $N=6$ curves nearly coincide at $K>12$, consistent with the sub-linear N exponent (0.28).

A. Straggler Robustness and APV Ablation

Fig. 5 tests heterogeneous FL: one device at $d=19$ m, $K-1=7$ at $[5,8]$ m. The proposed scheme allocates near-zero power to the straggler at tight ε ($<2\%$ share at $\varepsilon\le 0.07$), consistent with the water-filling structure of S2. FPA–MaxPow allocates full $P_{\max}$ regardless, widening the energy gap to ≈ 3.1 dBm at $\varepsilon=0.08$—larger than the ≈ 2.8 dBm homogeneous case.

Fig. 6 sweeps $C\in\{5,10,20,40,80,160\}$: converged energy plateaus beyond $C=40$ (gain to $C=160$ under 0.1 dBm), while runtime scales linearly with C, validating the $C=40$ design choice.

VI. Conclusion

We formulated the first joint port-repositioning and power-control problem for energy minimization in FAA-enhanced AirComp, proved monotone BCD convergence, and established convexity of $E^*(\varepsilon)$ via parametric convex programming. The scaling law $E^*\propto K^{1.34}N^{0.28}$ ($R^2=0.95$), over a $7\times3\times5$ grid with 500 MC runs per point, supports closed-form planning. At 5 GHz, up to 47% energy is saved with six FA ports. Future work includes multi-function AirComp and Byzantine-robust power control.

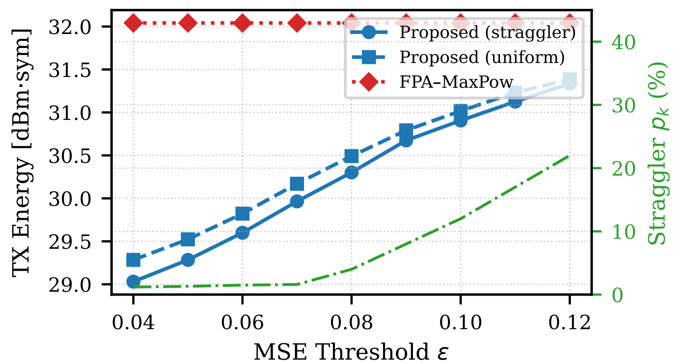

Fig. 5: Straggler robustness ($N=6$, $K=8$, one device at 19 m). Proposed scheme drives straggler power share below 2% at $\varepsilon\le 0.07$, outperforming FPA–MaxPow by ≈ 3.1 dBm.

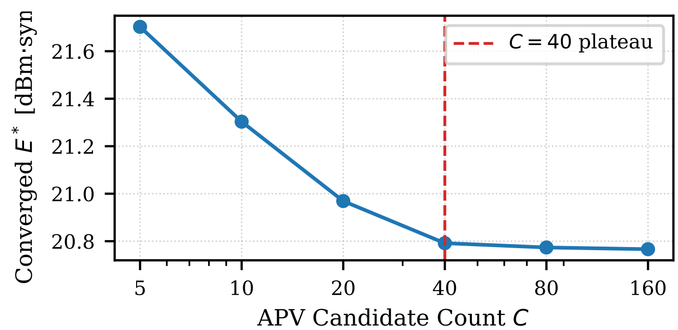

Fig. 6: Converged E^* vs. APV candidate count C. Energy plateaus beyond $C=40$; gains to $C=160$ are <0.1 dBm.

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